

CCNA 200-301: Class Notes — Day 3

Topic: Cisco Official Syllabus & Domain Breakdown

The **Cisco Certified Network Associate (CCNA 200-301)** exam tests a candidate's knowledge across six primary technology domains. Understanding the weight and key focus areas of each domain is essential for creating an effective study strategy.

Domain Overview & Weight Distribution

Domain #	Domain Name	Exam Weight (%)	Primary Focus Area
1.0	Network Fundamentals	20%	Hardware components, OSI/TCP-IP models, IPv4/IPv6, cabling
2.0	Network Access	20%	VLANs, Trunking, Spanning Tree (STP), Wireless (WLAN/WLC)
3.0	IP Connectivity	25%	Routing concepts, Static Routing, OSPFv2, Subnet routing
4.0	IP Services	10%	NAT, DHCP, DNS, NTP,

			SNMP, Syslog, QoS
5.0	Security Fundamentals	15%	ACLs, Layer 2 security, Port Security, VPNs, AAA, Wireless Security
6.0	Automation & Programmability	10%	SDN, REST APIs, Cisco DNA Center, JSON, Ansible/Puppet /Chef

Detailed Breakdown of Exam Domains

1.0 Network Fundamentals (20%)

This domain establishes the foundational bedrock of all enterprise communication.

- **Key Topics Covered:**

- **Network Components:** Roles and operations of Routers, Switches, Next-Gen Firewalls (NGFW), Access Points (APs), Wireless LAN Controllers (WLCs), and Endpoints.
- **Network Topologies:** Star, Mesh, Hybrid, 2-Tier/3-Tier Campus architectures, and Spine-Leaf designs.
- **Physical Layer Standards:** Copper cabling (Cat5e/Cat6), Fiber Optics (Single-Mode vs. Multi-Mode), and interface speeds.
- **Protocol Stack:** TCP/IP model vs. OSI model layers and encapsulation/decapsulation process.
- **IP Addressing & Subnetting:** IPv4 addressing, Variable Length Subnet Masking (VLSM), and IPv6 addressing types (Global Unicast,

Link-Local, Multicast).

2.0 Network Access (20%)

This domain covers switching concepts, VLAN topologies, and enterprise wireless access.

- **Key Topics Covered:**

- **VLAN Configuration & Trunking:** Creating VLANs, voice VLANs, access ports, and 802.1Q trunking protocols.
- **Layer 2 Discovery:** Cisco Discovery Protocol (CDP) and Link Layer Discovery Protocol (LLDP).
- **EtherChannel / Link Aggregation:** LACP (802.3ad) and static bundle configuration for port redundancy and increased bandwidth.
- **Spanning Tree Protocol (STP):** PVST+ and Rapid PVST+ operations, root bridge selection, port states, and PortFast/BPDU Guard features.
- **Wireless Architectures:** Autonomous vs. Controller-Based APs, Wireless LAN Controller (WLC) management, and SSID setup.

3.0 IP Connectivity (25%) — Highest Weighted Section

This domain focuses on routing logic, routing protocols, and how data packets move across networks.

- **Key Topics Covered:**

- **Routing Table Fundamentals:** Understanding Administrative Distance (AD), metrics, longest prefix match, and default gateways.
- **Static Routing:** Standard static routes, default routes (0.0.0.0/0), network routes, host routes, and floating static routes.
- **Dynamic Routing Protocols: OSPFv2 (Open Shortest Path First)** for single-area and multi-area setups (Neighbor adjacency, Router IDs, Designated Router/Backup Designated Router election).
- **First Hop Redundancy:** Understanding concepts of FHRP and High Availability gateways (HSRP).

4.0 IP Services (10%)

This domain highlights auxiliary protocols and services that support network operations and administration.

- **Key Topics Covered:**

- **Network Address Translation (NAT):** Static NAT, Dynamic NAT, and Port Address Translation (PAT / NAT Overload).
- **Time & Management Protocols:** Network Time Protocol (NTP), Syslog levels, and Simple Network Management Protocol (SNMP).
- **Core Infrastructure Services:** Dynamic Host Configuration Protocol (DHCP client/relay/server) and Domain Name System (DNS) resolution.
- **Quality of Service (QoS):** Per-hop behaviors, classification, marking (CoS/DSCP), queuing, and traffic shaping/policing concepts.
- **Remote Access:** Configuring secure SSH access and basic network management.

5.0 Security Fundamentals (15%)

This domain focuses on protecting network infrastructure against unauthorized access, attacks, and vulnerabilities.

- **Key Topics Covered:**

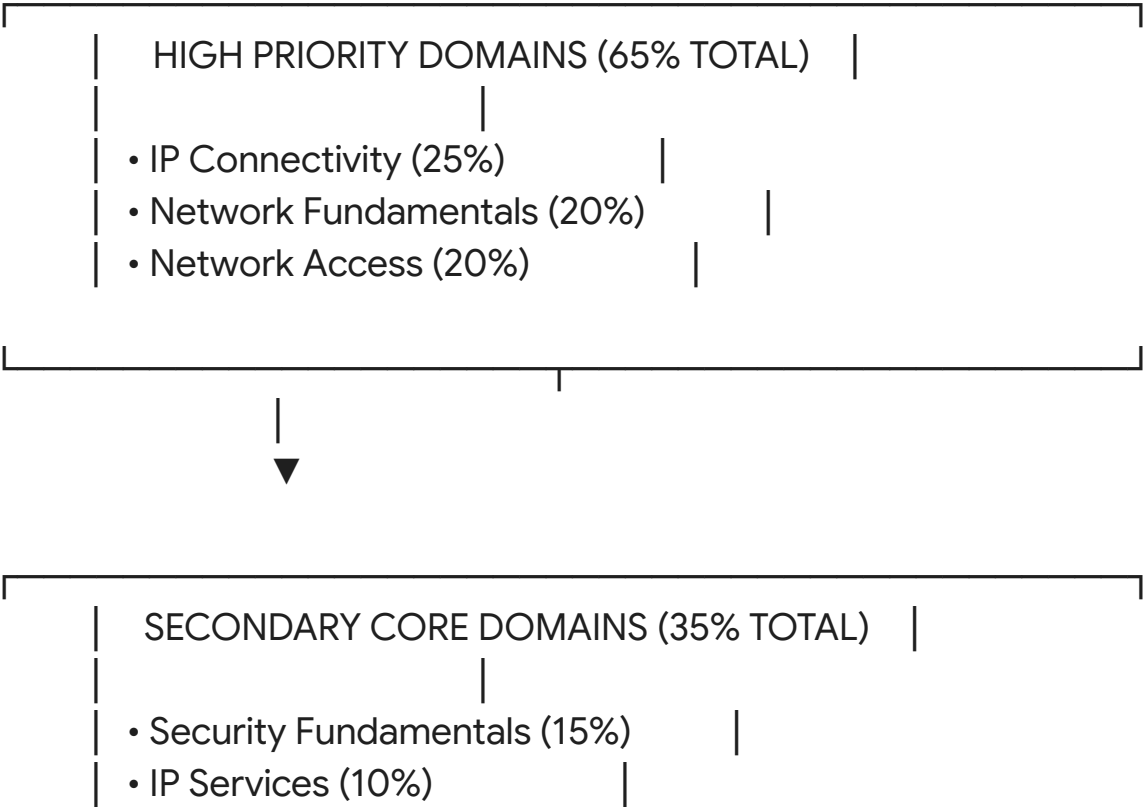
- **Security Concepts:** Threats, vulnerabilities, exploits, mitigation strategies, and defense-in-depth principles.
- **Access Control Lists (ACLs):** Standard and Extended IPv4 ACLs for traffic filtering and administrative access rules.
- **Layer 2 Infrastructure Security:** Port Security, Dynamic ARP Inspection (DAI), and DHCP Snooping.
- **Authentication & Access:** Local authentication, AAA (Authentication, Authorization, Accounting) using RADIUS and TACACS+.
- **Wireless & VPN Security:** WPA/WPA2/WPA3 protocols, 802.1X, site-to-site VPNs, and Remote Access VPNs.

6.0 Automation and Programmability (10%)

This domain prepares engineers for modern software-driven networks and cloud integrations.

- **Key Topics Covered:**
 - **Traditional vs. Controller-Based Networking:** Software-Defined Networking (SDN), control plane vs. data plane separation, and Cisco DNA Center / Catalyst Center.
 - **REST APIs:** HTTP verbs (GET, POST, PUT, DELETE), status codes, and API request mechanisms.
 - **Data Encoding Formats:** Reading and parsing **JSON**, **XML**, and **YAML** payloads.
 - **Configuration Management:** Overview of automation tools like **Ansible**, **Puppet**, and **Chef**.

Study Priority Strategy



| • Automation & Programmability (10%) |

Class Notes compiled from Day 3 Lecture outline.